HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2025)

Student No.: 18 Name: _____ Marks: / 93

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. If α, β and γ are non-zero constants such that $(\alpha+2\beta):(\beta+2\gamma):(\gamma+2\alpha)=4:9:5$, then $\alpha:\beta=$

   A. $2:5$.

   B. $5:2$.

   C. $128:149$.

   D. $149:128$.

(DSE-2025 PII Q11)

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2. The perimeter of the sector OMN is 12π cm, where O is the centre of the sector OMN. If the radius of the sector OMN is 3π cm, which of the following are true?

I. The area of the sector OMN is $9\pi^2$ cm^2.

II. The perimeter of $\triangle OMN$ is less than 35 cm.

III. The angle of the sector OMN is greater than $100°$.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

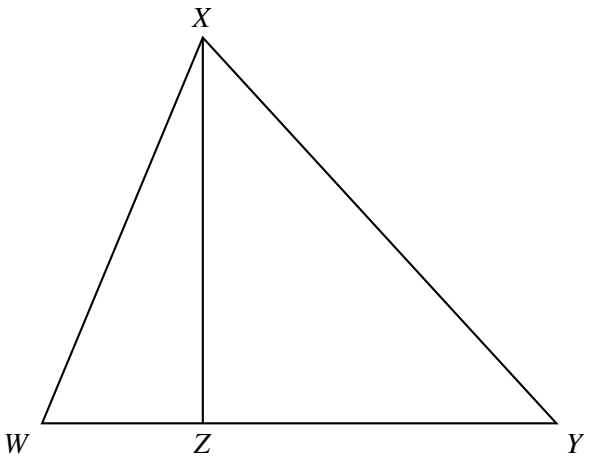
(DSE-2025 PII Q15)

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3. The figure shows the triangle WXY. Let Z be a point lying on WY such that $WY : XY = XY : YZ$. If $WX = 65$ cm, $WZ = 25$ cm and $XZ = 60$ cm, then $XY =$



A. 131 cm.

B. 144 cm.

C. 156 cm.

D. 169 cm.

(DSE-2025 PII Q18)

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4. It is given that $PQRS$ is a trapezium with $PQ \,//\, SR$. If $PS = 41\,\text{cm}$, $RS = 53\,\text{cm}$, $\angle PSR = 120^\circ$ and $\angle QRS = 150^\circ$, then $PQ =$

A. 82 cm.

B. 100 cm.

C. 106 cm.

D. 135 cm.

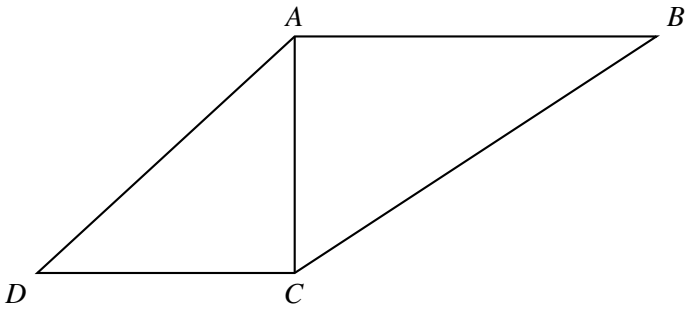
(DSE-2025 PII Q20)

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5. The figure shows the quadrilateral $ABCD$. If $\angle ABC + \angle ADC = \angle ACD = \angle BAC = 90°$, which of the following must be true?



A. $\tan\angle ACB = \dfrac{AB}{AD}$

B. $\tan\angle ACB = \dfrac{AB}{CD}$

C. $\tan\angle ADC = \dfrac{BC}{AD}$

D. $\tan\angle ADC = \dfrac{BC}{CD}$

(DSE-2025 PII Q23)

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6. The polar coordinates of the points X, Y and Z are $(1, 20^\circ)$, $(2, 80^\circ)$ and (r, θ) respectively, where $20^\circ < \theta < 80^\circ$. If $\triangle XYZ$ is an equilateral triangle, find r.

A. $\sqrt{3}$

B. $\sqrt{5}$

C. $\sqrt{7}$

D. $\sqrt{10}$

(DSE-2025 PII Q24)

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7. Denote the origin by O. The coordinates of the point A are $(a, 2a)$, where $a > 0$. Let P be a moving point in the rectangular coordinate plane such that $AP = OA$. The locus of P is a

A. point.

B. circle.

C. rhombus.

D. straight line.

(DSE-2025 PII Q25)

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8. The straight lines L_1 and L_2 are perpendicular to each other. The equations of L_1 and L_2 are $3x+4y-20=0$ and $mx+ny-20=0$ respectively, where m and n are constants. Suppose that L_1 and L_2 cut the x-axis at the points A and B respectively. Let C be the point of intersection of L_1 and L_2. It is given that C lies above the x-axis. If the area of $\triangle ABC$ is 6, find n.

A. -12

B. -9

C. 9

D. 12

(DSE-2025 PII Q26)

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9. The coordinates of the centre of the circle C are $(7,-5)$. If C cuts the x-axis at the points P and Q such that $PQ = 24$, then the equation of C is

A. $x^2+y^2-14x+10y-95=0$.

B. $x^2+y^2+14x-10y-95=0$.

C. $x^2+y^2-14x+10y-119=0$.

D. $x^2+y^2+14x-10y-119=0$.

(DSE-2025 PII Q27)

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10. $3E000000000000_{16} =$

A. $2^{16}+2^{15}+2^{14}+2^{13}+2^{12}$.

B. $2^{17}+2^{16}+2^{15}+2^{14}+2^{13}$.

C. $2^{52}+2^{51}+2^{50}+2^{49}+2^{48}$.

D. $2^{53}+2^{52}+2^{51}+2^{50}+2^{49}$.

(DSE-2025 PII Q31)

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11. On the same rectangular coordinate system, the graph of $y = \log_a x$ cuts the graph of $y = a^x$ and the x-axis at the points P and Q respectively, where a is a positive constant. Denote the origin by O. Which of the following are true?

I. $a < 1$

II. $OQ > a$

III. $\angle POQ = 45°$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2025 PII Q34)

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12. $i^9 + i^{10} + i^{11} + \cdots + i^{999} =$

A. -1.

B. 0.

C. 1.

D. i.

(DSE-2025 PII Q35)

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13. Consider the following system of inequalities:

$$\begin{cases} x \leq 11 \\ 4x + 5y - 19 \geq 0 \\ 7x - 6y + 11 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. If (x, y) is a point lying in D, then the greatest value of $8x - 6y + 11$ is

A. 1.

B. 11.

C. 15.

D. 129.

(DSE-2025 PII Q36)

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14. Let p, q and r be non-zero real numbers. If p, q, r is an arithmetic sequence, which of the following must be true?

I. 3^p, 3^q, 3^r is a geometric sequence.

II. $\frac{5}{p}$, $\frac{5}{q}$, $\frac{5}{r}$ is a geometric sequence.

III. $p-q$, $q-r$, $r-p$ is an arithmetic sequence.

A. I only

B. II only

C. I and III only

D. II and III only

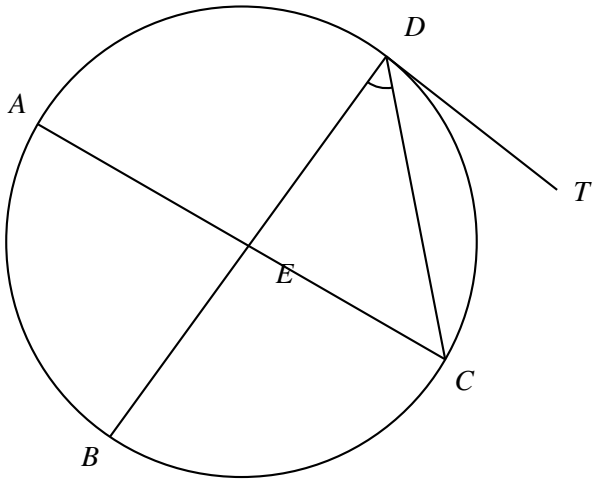
(DSE-2025 PII Q37)

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15. In the figure, AC is a diameter of the circle $ABCD$. Denote the point of intersection of AC and BD by E. It is given that TD is the tangent to the circle at D. If $\angle BEC = 96^\circ$ and $\angle CDT = 41^\circ$, then $\angle CDE =$



A. 47°.

B. 48°.

C. 52°.

D. 55°.

(DSE-2025 PII Q38)

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16. For $90° < \theta < 270°$, how many roots does the equation $\tan^3\theta = 2\tan\theta$ have?

A. 2

B. 3

C. 4

D. 5

(DSE-2025 PII Q39)

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17. $PQRS$ is a regular tetrahedron. Find the angle between PQ and $\triangle QRS$ correct to the nearest degree.

A. 35°

B. 55°

C. 60°

D. 71°

(DSE-2025 PII Q40)

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18. Denote the origin by O. The coordinates of the point U are $(20, 0)$. Let V be a point lying on the positive y-axis such that the x-coordinate of the in-centre of $\triangle OUV$ is 6. Find the area of $\triangle OUV$.

A. 70

B. 87

C. 210

D. 250

(DSE-2025 PII Q41)

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19. The coordinates of the point A are $(4, -2)$. A is rotated clockwise about the origin O through 90° to B.

(a) Write down the coordinates of B.

(b) Suppose that B is translated upwards by t units to the point C. If A, O and C are collinear, find t.

(3 marks) (DSE-2025 PI Q04)

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20. The selling price of a souvenir is $378. The souvenir is sold at a discount of 60% on its marked price. The marked price of the souvenir is 75% above its cost.

(a) Find the marked price of the souvenir.

(b) Find the cost of the souvenir.

(c) Determine whether there is a gain or a loss after selling the souvenir. Explain your answer.

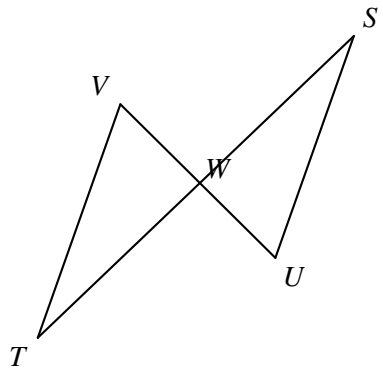
(5 marks) (DSE-2025 PI Q07)

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21. In the figure, ST and UV intersect at the point W. It is given that $SU \parallel VT$ and W is the mid-point of ST.



(a) Prove that $\triangle SUW \cong \triangle TVW$.

(b) Let X be a point lying on TW such that $\triangle SUW \sim \triangle VXW$. If $SU = 57\,\text{cm}$, $SW = 63\,\text{cm}$ and $WX = 7\,\text{cm}$, find the perimeter of $\triangle TVX$.

(5 marks) (DSE-2025 PI Q08)

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22. It is given that $p(x)$ partly varies as x and partly varies as x^2. Suppose that $p(7) = 56$ and $p(9) = 216$.

(a) Find $p(x)$.

(b) Let c be a real constant. Find the range of values of c such that the equation $p(x) = c$ has two distinct real roots.

(6 marks) (DSE-2025 PI Q11)

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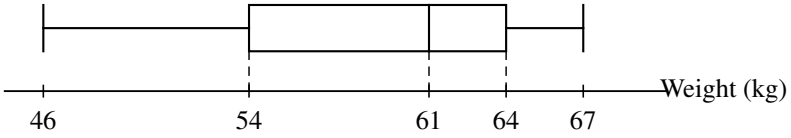
23. The stem-and-leaf diagram below shows the distribution of the weights (in kg) of some athletes before training.

Stem (tens)	Leaf (units)
4	5 6 8
5	w w 6 7
6	3 4 5 6 9 9
7	0 2
8	7

The difference of the range and the inter-quartile range of the above distribution is 25 kg.

(a) Write down the range of the above distribution. Hence, find w.

(b) The box-and-whisker diagram below shows the distribution of the weights (in kg) of the athletes after the training.



(i) Find the change in the upper quartile of the distribution due to the training.

(ii) Is the distribution of the weights of the athletes after the training less dispersed than that before the training? Explain your answer.

(7 marks) (DSE-2025 PI Q12)

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24. It is given that $\begin{cases} \log_3 x + \log_3 y = 9 \\ \log_x 81 - \log_y 9 = 1 \end{cases}$, where $0 < x < y$.

(a) Let $u = \log_3 y$. Prove that $u^2 - 3u - 18 = 0$.

(b) Find x.

(4 marks) (DSE-2025 PI Q16)

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25. Let $T(n)$ be the nth term of an arithmetic sequence such that $T(1) \neq T(2)$ and $T(47) = 456$. It is given that $T(9)$, $T(47)$, $T(199)$ is a geometric sequence.

(a) Find $T(1)$.

(b) Find the least value of n such that the sum of the first n terms of the arithmetic sequence is greater than 10^6.

(6 marks) (DSE-2025 PI Q17)

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26. Let $g(x) = 3x^2 - 6kx + 24x + 3k^2 - 24k + 55$, where k is a positive constant. Denote the vertex of the graph of $y = g(x)$ by R.

(a) Using the method of completing the square, express the coordinates of R in terms of k.

(b) The graph of $y = h(x)$ is obtained by translating the graph of $y = g(x)$ rightwards by 6 units and then downwards by 10 units. Denote the vertex of the graph of $y = h(x)$ by S. Let T be a point such that the orthocentre of $\triangle RST$ is the origin.

(i) Express the coordinates of T in terms of k.

(ii) The coordinates of the point U are $(-5, 5)$. Let V be a point lying on RS such that TV is perpendicular to RS. It is given that S, T, U and V are concyclic. Find k.

(9 marks) (DSE-2025 PI Q18)

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27. The equation of the straight line L is $4x-3y+83=0$. The equation of the circle C is $x^2+y^2+ax-2y+b=0$, where a and b are positive constants. It is given that C passes through the point $(-10,9)$ and L is a tangent to C.

(a) Prove that the equation $25x^2+(9a+640)x+(90a+4924)=0$ has equal roots. Hence, find a and b.

(b) The coordinates of the point P are $(-20,1)$. Let Q be a point lying below L such that PQ is tangent to C and $PQ=25$. Denote the centre of C by I.

(i) Find $\cos\angle IPQ$.

(ii) Find, in surd form, the distance between I and Q.

(iii) Let R be a point lying on L such that C is the inscribed circle of $\triangle PQR$. Denote the radius of the circumcircle of $\triangle PQR$ by r. Someone claims that $r>PQ$. Is the claim correct? Explain your answer.

(12 marks) (DSE-2025 PI Q19)

I can finish _____ questions

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